

CONCISE DESCRIPTION OF RELEVANCE: THIRD-PARTY SUBMISSION UNDER 37 CFR § 1.290

U.S. Application No. 19/318,450 (Publication No. US 2026/0070232 A1)

Document 6: U.S. Patent Application Publication No. US 2024/0087784 A1 (“Document 6”)

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Title: Magnetic coupling device

This document discloses a magnetic coupling device, which may include sensors, retractable pins, stationary pins, probes, and/or additional tools, for magnetically coupling to a ferromagnetic workpiece. Its potential relevance to the claims of the above-identified application is set forth below.

Claim language (US 2026/0070232 A1)	Relevant disclosure of this document
Claim 1: “A magnetic gripper for gripping a ferromagnetic workpiece, said magnetic gripper comprising”	The reference discloses a magnetic coupling device for magnetically coupling to a ferromagnetic workpiece, comprising a housing. “A magnetic coupling device for magnetically coupling to a ferromagnetic workpiece, comprising: a housing ...” • Claim 1
Claim 1: “a magnet which is arranged in the housing and which is displaceable along a displacement direction between a gripping position for gripping the ferromagnetic workpiece and a release position for releasing the ferromagnetic workpiece”	The reference discloses a switchable magnetic flux source supported by the housing, including a plurality of permanent magnets, where a first permanent magnet and a second permanent magnet are movable relative to each other. “a switchable magnetic flux source supported by the housing including a plurality of permanent magnets, the plurality of permanent magnets including a first permanent magnet and a second permanent magnet movable relative to the first permanent magnet ...” • Claim 1
Claim 1: “a number of pole shoes attached to the housing, wherein each pole shoe has a workpiece contact surface and is designed to guide a magnetic field portion of the magnet to the workpiece contact surface”	The reference discloses a plurality of workpiece engagement surfaces supported by the housing and magnetically coupled to the switchable magnetic flux source, which are adapted to contact the ferromagnetic workpiece. “a plurality of workpiece engagement surfaces supported by the housing and magnetically coupled to the switchable magnetic flux source, the plurality of workpiece engagement surfaces adapted to contact the ferromagnetic workpiece ...” • Claim 1

Claim language (US 2026/0070232 A1)	Relevant disclosure of this document
Claim 6 (the active structure has a recess between the first projection and the second projection)	The reference discloses a channel located between the first north pole portion and the first south pole portion. “wherein a channel is located between the first north pole portion and the first south pole portion.” • Claim 1
Claim 8 (The magnetic gripper according to claim 1, wherein the number of pole shoes includes a first pole shoe and a second pole shoe)	The reference discloses a first north pole portion having a first workpiece engagement surface, a second north pole portion having a second workpiece engagement surface, a first south pole portion having a third workpiece engagement surface, and a second south pole portion. “a first north pole portion having a first workpiece engagement surface, a second north pole portion having a second workpiece engagement surface, a first south pole portion having a third workpiece engagement surface, and a second south pole portion having ...” • Claim 1
Claim 10 (The magnetic gripper according to claim 1, wherein the magnetic gripper has a position sensor for detecting the gripping position and/or the release position)	The reference discloses that the magnetic coupling devices may include sensors. “The magnetic coupling devices may include sensors, retractable pins, stationary pins, probes, and/or additional tools.” • Abstract
Claim 11 (The magnetic gripper according to one of claim 1, wherein the housing has a receptacle for receiving a workpiece sensor and/or a supply line)	The reference discloses that the magnetic coupling devices may include sensors. “The magnetic coupling devices may include sensors, retractable pins, stationary pins, probes, and/or additional tools.” • Abstract
Claim 14 (a magnetic gripper according to claim 5, and the ferromagnetic workpiece)	The reference discloses a magnetic coupling device for magnetically coupling to a ferromagnetic workpiece. “A magnetic coupling device for magnetically coupling to a ferromagnetic workpiece ...” • Claim 1